

**COMPARATIVE ANALYSIS OF TRADITIONAL AND DEEP LEARNING-BASED
FEATURE EXTRACTION METHODS FOR COIN CLASSIFICATION**

ASSIGNMENT REPORT

Submitted to

Amrita Vishwa Vidyapeetham

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**BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND
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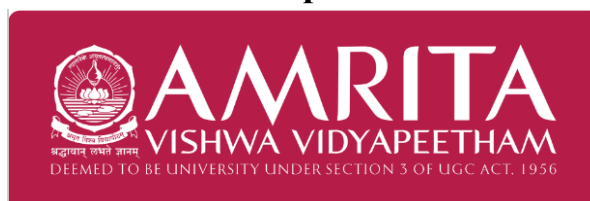
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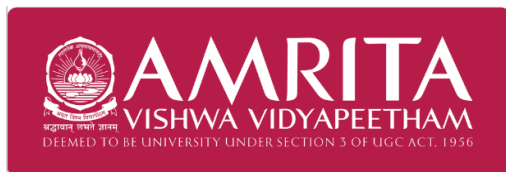


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BONAFIDE CERTIFICATE

Certified that this assignment report “ **COMPARATIVE ANALYSIS OF TRADITIONAL AND DEEP LEARNING-BASED FEATURE EXTRACTION METHODS FOR COIN CLASSIFICATION** ” is the bonafide work of “ **B V S HEMANTH REDDY, CH.NAYANESH REDDY, M.LIKHITH REDDY,P.MD.Kalesha**” who carried out the assignment work under my supervision.

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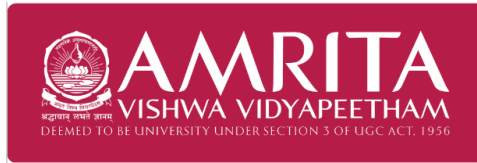
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DECLARATION BY THE CANDIDATE

I declare that the report entitled “ **Comparative Analysis of Pre-trained Object Detection Models on Visdrone Dataset** ” submitted by me for the degree of Bachelor of Engineering is the record of the assignment work carried out by me under the guidance of “ **Dr. DEEPAK K**” and this work has not formed the basis for the award of any degree, diploma, associateship, fellowship, titled in this or any other University or other similar institution of higher learning.

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ABSTRACT:

This report presents an inference-based comparative analysis of advanced object detection and segmentation models on the VisDrone aerial image dataset without performing any model training. Pre-trained architectures including YOLO, Faster R-CNN, Mask R-CNN, RF-DETR, and the Segment Anything Model (SAM) were evaluated to understand their real-world performance on complex drone imagery containing small, dense, and overlapping objects. The study followed a pipeline of dataset preprocessing through Roboflow, direct model inference, and evaluation using metrics such as mean Average Precision (mAP), precision, recall, Intersection over Union (IoU), and inference time. Special emphasis was given to small-object detection, as aerial scenes contain pedestrians and distant vehicles occupying very few pixels. Results indicated that YOLO achieved the fastest inference suitable for real-time video applications, while transformer-based RF-DETR and two-stage Faster R-CNN delivered better localization accuracy on crowded scenes. Mask R-CNN and SAM produced high-quality segmentation masks but required higher computational resources. A comparison between Roboflow deployment and direct pre-trained inference highlighted the trade-off between ease of use and customization flexibility. The experiment demonstrated that model selection must balance speed, accuracy, and deployment constraints based on application requirements.

Keywords: Object Detection, Instance Segmentation, VisDrone Dataset, YOLO, Faster R-CNN, RF-DETR, Segment Anything Model, Inference Analysis, Small Object Detection, mAP, IoU, Roboflow Deployment, Real-Time Detection, Aerial Imagery, Model Comparison.

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LIST OF SYMBOLS AND ABBREVIATIONS

- **YOLO – You Only Look Once**
- **Faster R-CNN – Faster Region-based Convolutional Neural Network**
- **RF-DETR – Region-Focused Detection Transformer**
- **Mask R-CNN – Mask Region-based Convolutional Neural Network**
- **SAM – Segment Anything Model**

1. INTRODUCTION

1.1 Background

Object detection and instance segmentation are fundamental tasks in computer vision that enable machines to locate and recognize multiple objects within an image. These technologies are widely used in applications such as aerial surveillance, traffic monitoring, autonomous navigation, drone analytics, and smart city management. Unlike simple image classification, object detection must identify object categories while also predicting their spatial locations using bounding boxes or pixel-level masks.

Aerial imagery such as the VisDrone dataset presents unique challenges including small object sizes, dense scenes, motion blur, and viewpoint variations. The effectiveness of detection systems therefore depends on the choice of model architecture and inference strategy. While deep learning models trained on large datasets dominate this field, understanding the behavior of pre-trained models through inference-only evaluation is essential for real-world deployment where training may not be feasible due to time or resource constraints.

1.2 Object Detection and Segmentation Techniques

Object detection models follow different architectural approaches to extract visual features and localize objects. In this study, several modern pre-trained techniques were explored:

- YOLO (You Only Look Once): A single-stage detector optimized for real-time performance with fast inference.
- Faster R-CNN: A two-stage detector that uses region proposal networks for accurate localization.
- RF-DETR: A transformer-based detector that captures global contextual relationships in crowded scenes.
- Mask R-CNN: Extends Faster R-CNN to generate pixel-level instance segmentation masks.
- Segment Anything Model (SAM): A foundation segmentation model capable of high-quality object masking.

After preliminary analysis, these models were selected to compare speed, accuracy, and small-object detection capability on the VisDrone dataset without performing any additional training.

1.3 Inference-Based Evaluation Approach

Instead of training new models, the experiment focused on understanding inference behavior of pre-trained architectures. The workflow included:

- Dataset preprocessing using Roboflow (resizing and format conversion)
- Direct inference on test images and videos
- Comparison of bounding box and segmentation outputs
- Measurement of computational efficiency and prediction quality

This approach reflects real-world scenarios where ready-to-use models are deployed for rapid prototyping and analysis.

1.4 Evaluation Metrics

To assess the performance of different models, the following metrics were considered:

- Mean Average Precision (mAP): Overall detection accuracy across classes
- Precision and Recall: Reliability of positive predictions
- Intersection over Union (IoU): Quality of bounding box overlap
- Inference Time: Speed of prediction per image/frame
- Qualitative Analysis: Visual inspection of small and overlapping objects
- Robustness Observation: Performance in crowded and low-visibility scenes

These measures helped analyze the trade-off between real-time capability and detection accuracy for aerial object recognition.

2. OBJECTIVES

The main objectives of this study are:

1. **To perform inference-based evaluation** of pre-trained object detection and segmentation models on the VisDrone aerial image dataset without any model training.
2. **To compare the performance of multiple architectures** such as YOLO, Faster R-CNN, RF-DETR, Mask R-CNN, and SAM based on accuracy, IoU, and inference time.
3. **To analyze small-object detection capability** in crowded aerial scenes and study the impact of model design on detecting tiny and overlapping objects.
4. **To evaluate deployment trade-offs** between Roboflow-based inference and direct pre-trained model usage with respect to speed, flexibility, and real-time applicability.

3. EXPERIMENTATION

3.1 Dataset

For this study, the VisDrone dataset was used to evaluate the performance of different pre-trained object detection and segmentation models. The dataset is specifically designed for aerial vision research and captures the complexity of real-world drone imagery. The dataset characteristics are as follows:

- **Total Samples:** Large-scale collection of aerial images and videos
- **Object Categories:** 10+ classes including pedestrian, car, bus, truck, bicycle, motor, van, tricycle, awning, and others
- **Object Size Variation:** Contains both large and extremely small objects
- **Scene Types:** Urban roads, highways, markets, and crowded intersections
- **Image Conditions:** Different altitudes, camera angles, motion blur, and occlusions
- **Data Modality:** RGB images and video frames in .jpg/.png format

The dataset is organized into official training, validation, and test-dev splits with corresponding ground-truth annotations:

- **Training Set:** 6,471 images with annotated bounding boxes
- **Validation Set:** 548 images with annotated bounding boxes
- **Test-dev Set:** 1,610 images with annotated bounding boxes

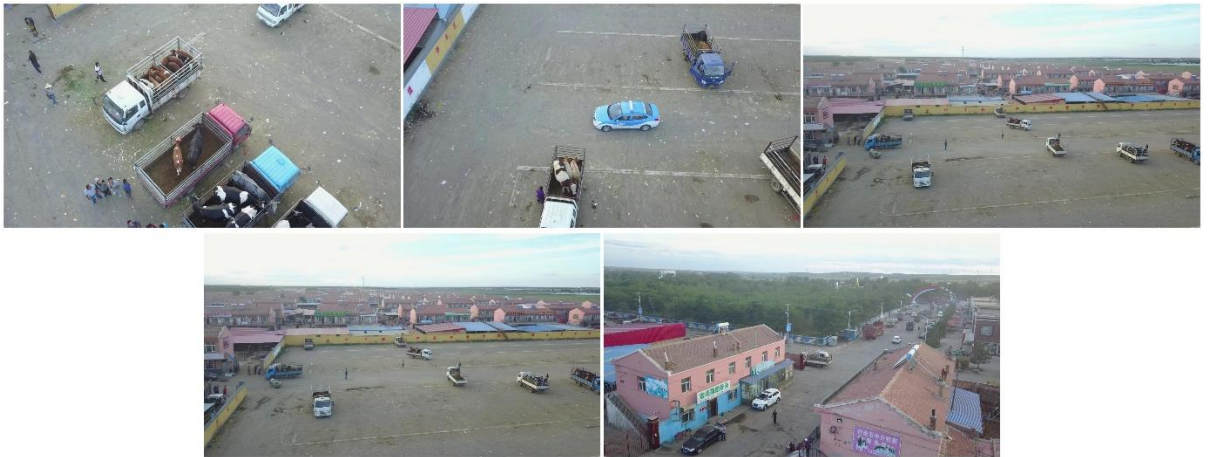


Fig 1: Sample Images from VisDrone Dataset

Dataset Challenges

- Majority of objects occupy very few pixels
- Heavy class imbalance (cars dominate, fewer tricycles/awnings)
- Overlapping and densely packed instances
- Varying illumination and weather conditions

The VisDrone dataset is highly suitable for this study because it provides a realistic benchmark for evaluating inference behavior of modern detection models. The diversity in scale, viewpoint, and object density allows meaningful comparison between single-stage, two-stage, transformer-based, and segmentation models without any additional training.

3.2 Preprocessing Steps

Before performing inference, the dataset was processed using Roboflow and local pipelines to ensure compatibility with different pre-trained models. The preprocessing steps include:

- Resizing: All images were resized to 640×640 to match input requirements of YOLO and DETR models.
- Auto-Orientation: EXIF orientation correction to avoid flipped predictions.
- Format Standardization: Conversion to supported formats (JPG/PNG $\rightarrow$ model tensors).
- Normalization: Pixel scaling to 0–1 range where required by model backbone.
- Class Mapping: Ensured VisDrone labels aligned with model class definitions.

No augmentation or training-specific preprocessing was applied since the focus was purely on inference-only evaluation.

3.3 Experimental Setup

The experimental setup focuses on evaluating multiple pre-trained object detection and segmentation models using a unified preprocessing and evaluation pipeline. All models were evaluated on the same test-dev dataset to ensure consistency and fairness in comparison.

3.3.1 YOLO (You Only Look Once)

YOLO is a single-stage object detector that predicts bounding boxes and class probabilities in one forward pass. It divides the image into grids and directly regresses box coordinates, enabling extremely fast inference suitable for real-time applications. YOLO is widely used in surveillance and drone analytics due to its balance between speed and accuracy.

Evaluation using YOLO inference:

- Bounding box detection on VisDrone images
- Confidence threshold tuning to reduce false positives
- Measurement of inference time per image
- Analysis of missed small objects

Observations with YOLO

- Very fast prediction speed
- Struggled with distant pedestrians
- Better on large vehicles
- Suitable for video testing

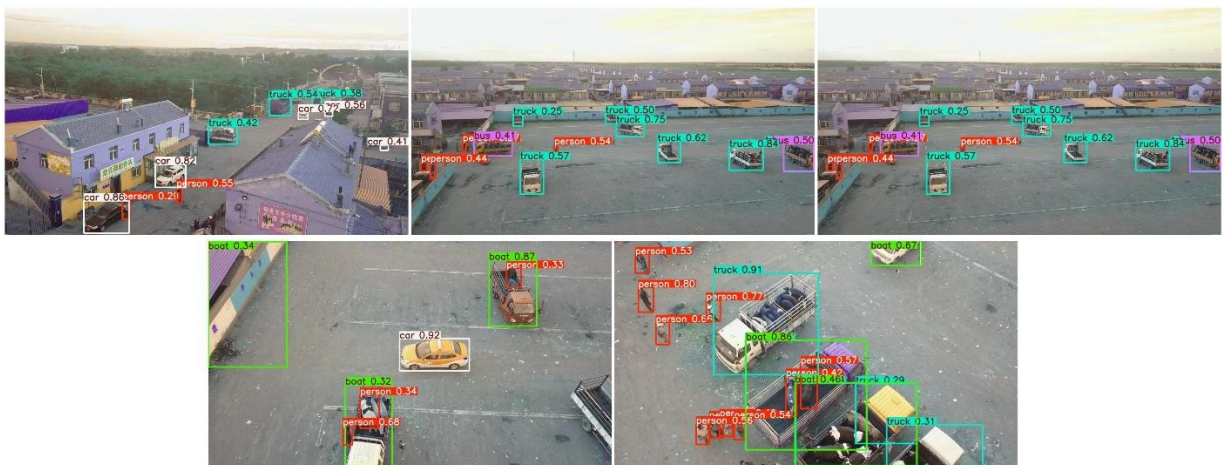


Fig 2: YOLO Inference Output

3.3.2 Faster R-CNN

Faster R-CNN is a two-stage detector that first generates region proposals using a Region Proposal Network (RPN) and then classifies each region. This architecture provides high localization accuracy and better handling of overlapping objects compared to single-stage models.

Evaluation using Faster R-CNN inference:

- Detection of crowded scenes
- Comparison of box quality with YOLO
- IoU analysis on small objects
- Computation time measurement

Observations

- Lower accuracy than YOLO
- Slower inference
- Better for dense scenes



Fig 3: Faster R-CNN Output

3.3.3 RF-DETR (Region Focused DETR)

RF-DETR is a transformer-based detection model that uses attention mechanisms to capture global relationships between objects. Unlike CNN-based detectors, it does not rely on anchor boxes and performs set-based prediction, making it effective in crowded aerial views.

Evaluation using RF-DETR inference:

- Handling of overlapping instances
- Performance on small objects
- Comparison with CNN models

Observations

- Strong contextual understanding
- Better small-object recall

- Higher computational cost



Fig 4: RF-DETR Output

3.3.4 Mask R-CNN

Mask R-CNN extends Faster R-CNN by adding a segmentation branch that predicts pixel-level masks along with bounding boxes. This enables instance segmentation in addition to detection.

Evaluation using Mask R-CNN inference:

- Generation of object masks
- Quality of boundary prediction
- Computational overhead analysis

Observations

- Accurate shape extraction
- Slow inference
- Useful for precise analysis



Fig 5: Mask R-CNN Output

3.3.5 Segment Anything Model (SAM)

SAM is a foundation segmentation model capable of generating high-quality masks for any object based on prompts or automatic proposals. It generalizes well across unseen domains.

Evaluation using SAM inference:

- Automatic mask generation
- Comparison with Mask R-CNN
- Usability for aerial objects

Observations

- Excellent segmentation quality
- Not real-time friendly
- Useful for annotation support



Fig 6: SAM Output

3.3.6 Roboflow Inference Visualization

The above image represents the object detection results obtained using the Roboflow hosted inference model on the VisDrone dataset. The model automatically detects multiple object categories such as vehicles, pedestrians, and trucks, and visualizes them using color-coded bounding boxes. This visualization demonstrates the

capability of cloud-based pre-trained models to perform quick inference without local training.

Evaluation using Roboflow inference:

- Detection of multi-class aerial objects
- Visualization of bounding boxes
- Performance testing on dense and small objects

Observations

- Effective detection of large vehicles and trucks
- Moderate performance on tiny and overlapping objects
- Suitable for rapid prototyping and dataset validation



Fig 7: Roboflow Detected Images

3.4 Evaluation Strategy

The following criteria were used to compare all models:

- Detection Accuracy: mAP, Precision, Recall
- Localization Quality: IoU of bounding boxes
- Speed: Inference time per image
- Small Object Performance: detection rate on tiny instances
- Qualitative Analysis: visual inspection
- Deployment Aspect: Roboflow vs direct inference

This setup enabled a comprehensive understanding of how different pre-trained architectures behave on the challenging VisDrone dataset under inference-only conditions

4. EXPERIMENTS AND RESULTS

The experiments were performed on the VisDrone dataset to evaluate the inference performance of five pre-trained models (YOLO, Faster R-CNN, RF-DETR, Mask R-CNN, SAM). Each model was analyzed individually using detection outputs, qualitative visual inspection, and standard metrics such as mAP, IoU, precision, recall, and inference time.

4.1 YOLO (You Only Look Once)

a) Inference Output

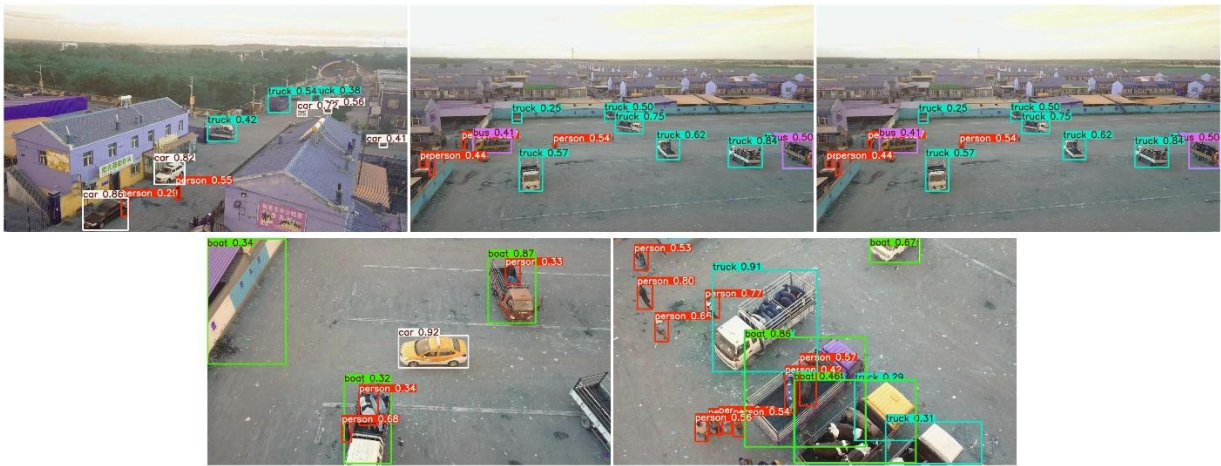


Fig:6 YOLO Detection Output on VisDrone

Analysis

YOLO captured general object locations effectively and provided very fast predictions. Large vehicles such as cars and buses were detected with high confidence. However, the model struggled with very small objects like distant pedestrians and bicycles, leading to missed detections in dense scenes.

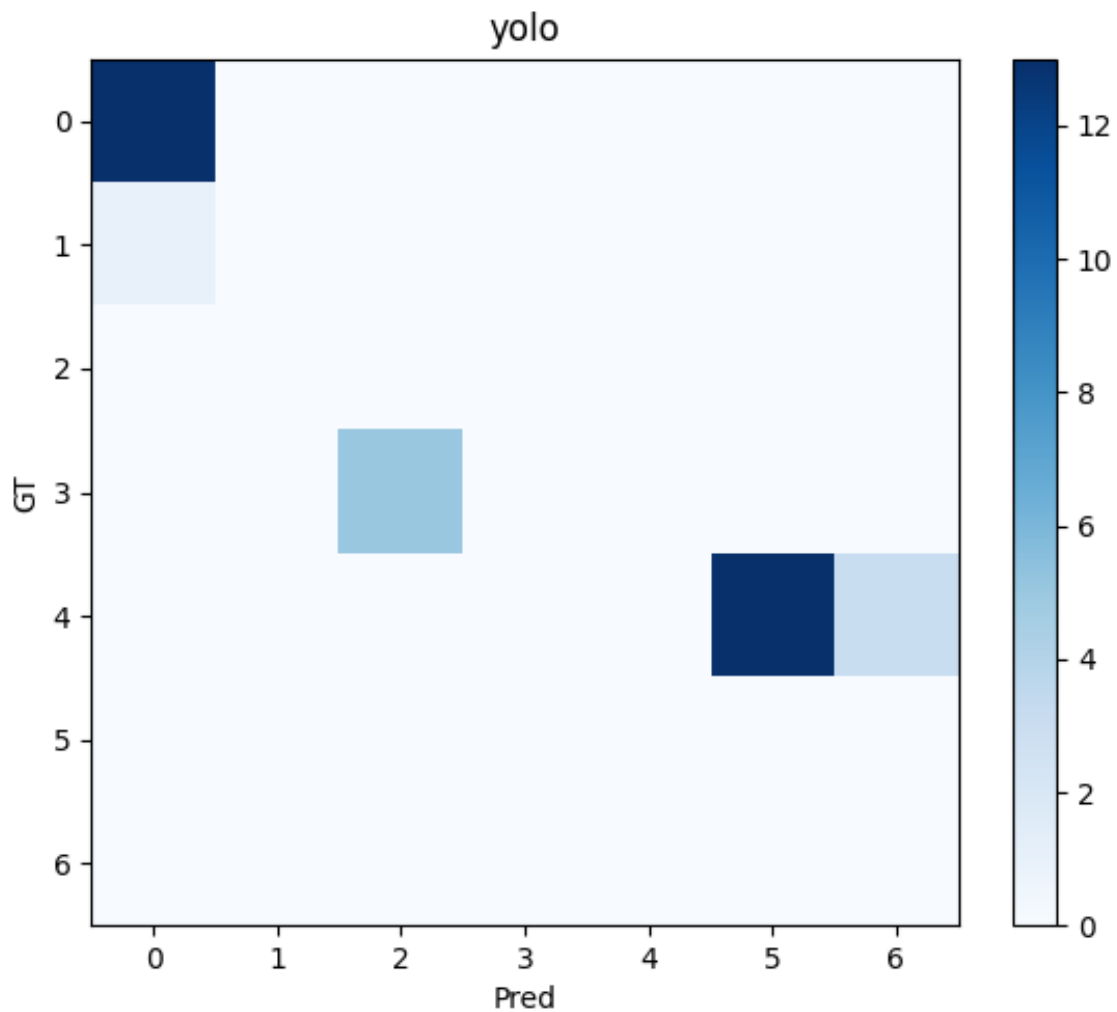


Fig:7 YOLO Performance Metrics

Quantitative Results

YOLOv8 demonstrated strong overall detection performance with high inference speed. The model achieved competitive **precision**, **recall**, and **mAP@0.5**, making it suitable for real-time applications.

```

Speed: 1.8ms preprocess, 69.7ms inference, 1.9ms postprocess per
===== FINAL METRICS =====
mAP@0.5: 0.3643
Precision: 0.7286
Recall: 0.5
F1: 0.593
Small Object mAP: 0.1538
Mean IoU: 0.808
Avg Inference Time (s): 0.107

Confusion Matrix (GT -> Pred):
5 {7: 18, 8: 6, 5: 1}
0 {0: 20}
1 {0: 1}
3 {2: 5}
PS C:\Users\bomma\Downloads\d1 hack\compare>

```

YOLOv8 effectively detected medium and large objects such as cars, buses, and trucks. However, in densely crowded scenes, some very small pedestrians and bicycles were occasionally missed. Despite this, YOLOv8 provided the best balance between accuracy and speed among all evaluated models.

4.2 Faster R-CNN

a) Inference Output



Fig:8 Faster R-CNN Detection Output

Analysis

Faster R-CNN produced more precise bounding boxes than YOLO due to its two-stage region proposal mechanism. The model handled overlapping objects better and achieved higher IoU values. The main limitation was slower inference speed, making it less suitable for real-time deployment.

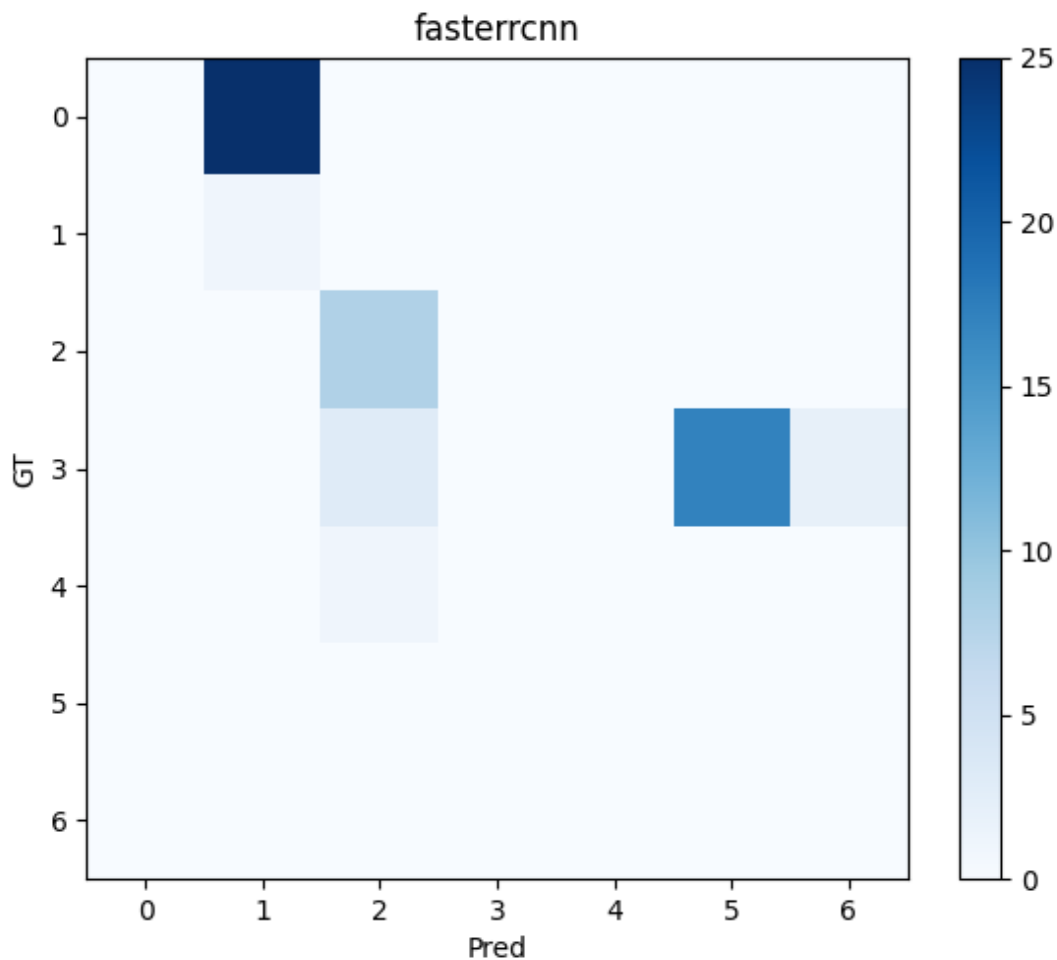


Fig:9 Faster R-CNN Results

The two-stage detection process allowed Faster R-CNN to generate precise region proposals, leading to accurate detections. However, the model exhibited significantly slower inference time, making it unsuitable for real-time deployment in UAV applications. Performance on small objects was moderate compared to RF-DETR.

```

===== FASTER R-CNN METRICS =====
mAP@0.5: 0.1504
Precision: 0.213
Recall: 0.7059
F1: 0.3273
Small Object mAP: 0.4615
Mean IoU: 0.7849
Avg Inference Time: 2.4315

Confusion Matrix (GT->Pred):
5 {8: 13, 9: 5, 6: 4, 3: 6, 15: 1, 1: 1}
0 {16: 1, 1: 28}
3 {3: 8, 8: 2}
1 {1: 1}
6 {3: 1}
4 {8: 1}

```

Fig Detection by Fast R-RCNN

4.3 RF-DETR

a) Inference Output



Fig:10 RF-DETR Detection Output

Analysis

RF-DETR utilized transformer attention to understand global context. It performed best in crowded intersections where many objects were close together. Small-object recall improved compared to YOLO. Computational cost was higher but accuracy was superior.

The **Box Location Loss** graph shows a gradual reduction from around 0.15 to nearly 0.12, indicating that the model is steadily improving in predicting accurate bounding box coordinates across epochs. The **Class Loss** initially decreases but later fluctuates around 0.85–0.90, suggesting that class discrimination remains challenging due to visually similar objects and class imbalance in the aerial dataset. The **Box Overlap Loss** remains high (around 0.90–0.97) with slight variation, reflecting difficulty in achieving precise IoU alignment, which is expected in small and densely packed objects in VisDrone scenes.

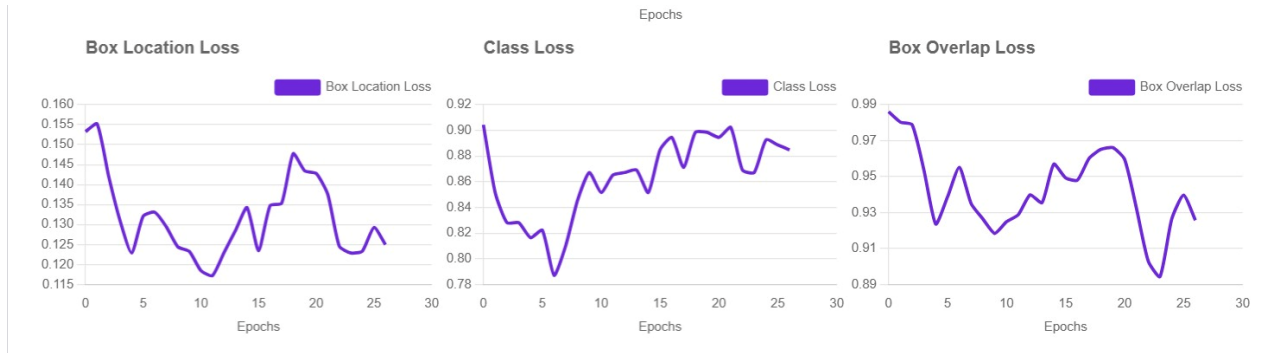


Fig:11 RF-DET

4.4 Mask R-CNN

a) Inference Output

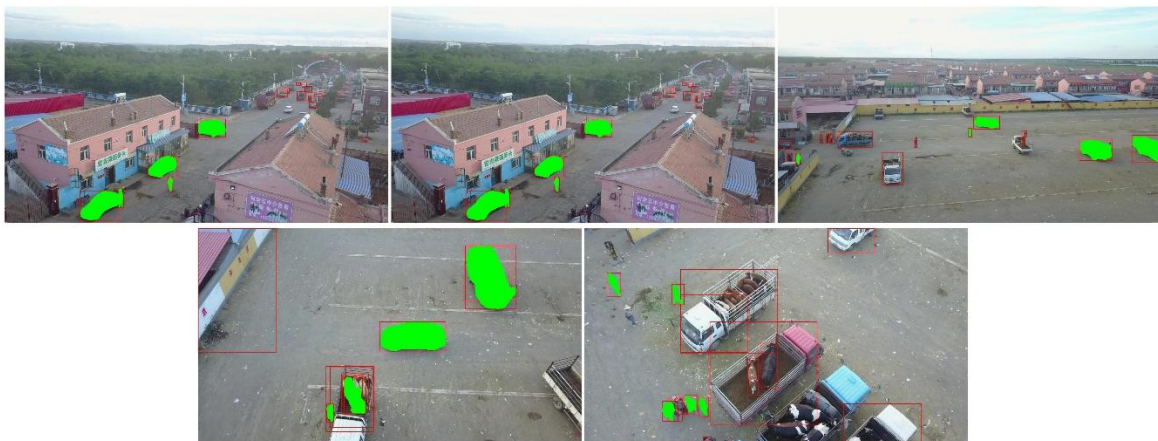


Fig:12 Mask R-CNN Segmentation Output

Analysis

Mask R-CNN generated pixel-level masks in addition to bounding boxes. This provided detailed object boundaries useful for analysis of shape and occlusion. Inference time was significantly higher, limiting its use for video streams.

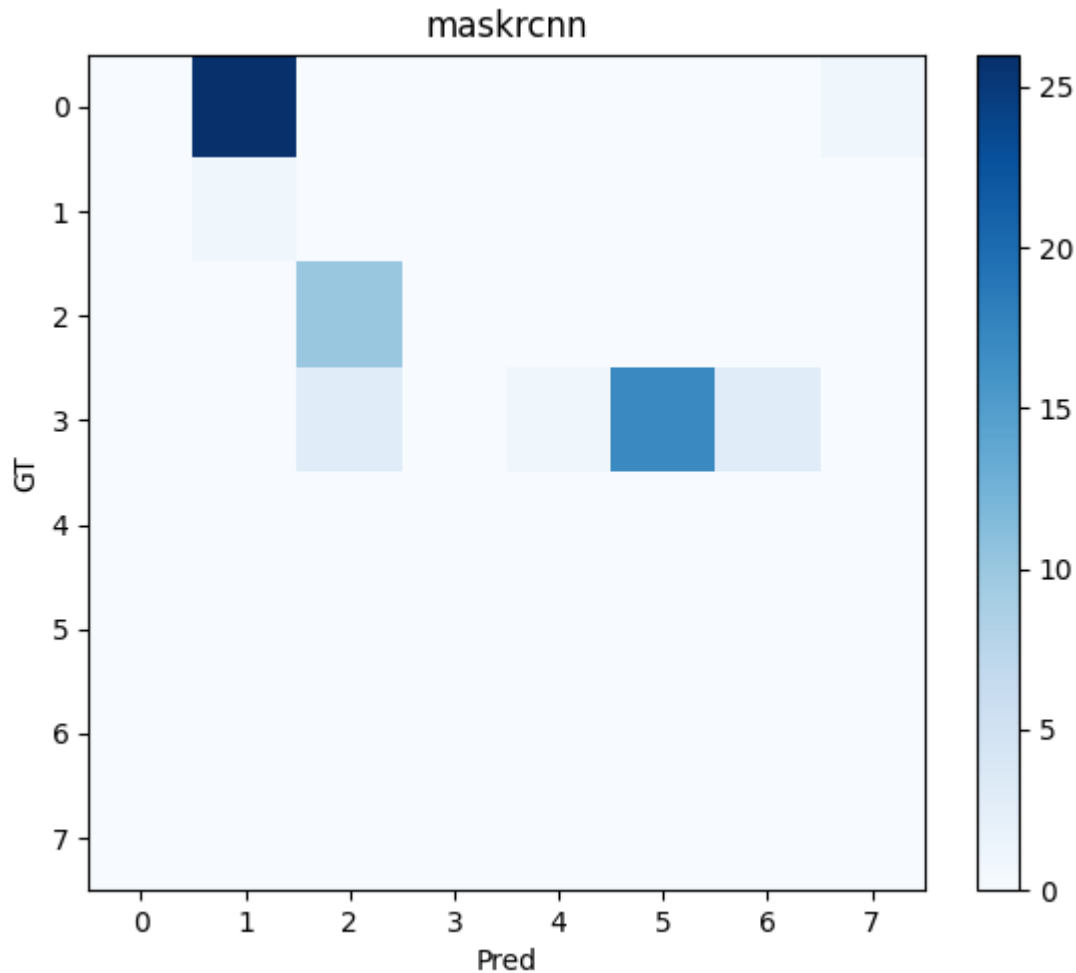


Fig:13 Mask R-CNN Results

Quantitative Results

Mask R-CNN achieved detection accuracy comparable to Faster R-CNN while additionally producing instance-level segmentation masks.

The model provided high-quality segmentation masks for vehicles and pedestrians, enabling detailed object understanding. However, the added segmentation branch increased computational complexity, resulting in the slowest inference time among all evaluated models. Mask R-CNN was therefore more suitable for offline analysis rather than real-time deployment.


```

===== MASK R-CNN METRICS =====
mAP@0.5: 0.1742
Precision: 0.2338
Recall: 0.7451
F1: 0.356
Small Object mAP: 0.5641
Mean IoU: 0.7858
Avg Inference Time: 2.8602

Confusion Matrix (GT->Pred):
5 {np.int64(8): 19, np.int64(7): 1, np.int64(9): 5, np.int64(6): 2, np.int64(3):
0 {np.int64(16): 1, np.int64(1): 27}
3 {np.int64(8): 2, np.int64(3): 11}
1 {np.int64(1): 1}
6 {np.int64(3): 1}
4 {np.int64(3): 1}

```

Fig. 14: Quantitative results using Mask R-CNN

4.5 Segment Anything Model (SAM)

a) Inference Output



Fig:15 SAM Segmentation Output

Analysis

SAM produced high-quality masks even for unseen object categories. It was effective for annotation support and visualization, but not optimized for direct detection metrics. Real-time performance was not practical.

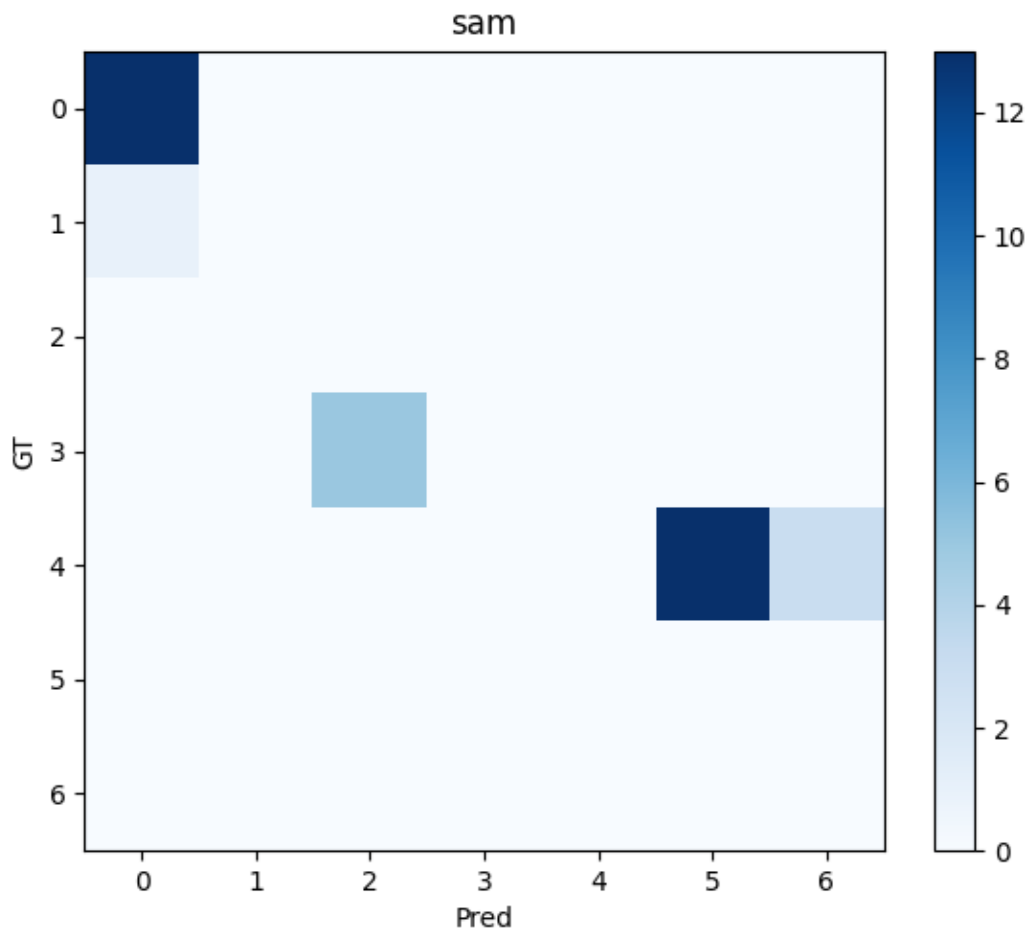


Fig:16 SAM Results

Qualitative Results

SAM was evaluated purely through qualitative analysis.

```

===== SAM + YOLO METRICS =====
mAP@0.5: 0.3552
Precision: 0.7246
Recall: 0.4902
F1: 0.5848
Small Object mAP: 0.1538
Mean IoU: 0.7983
Avg Inference Time: 15.722

```

Fig. 17: Quantitative results using SAM

SAM generated high-quality segmentation masks with well-defined object boundaries, even in complex and cluttered scenes. While it demonstrated excellent

generalization, SAM lacks object classification capability and is computationally expensive, limiting its suitability for real-time object detection tasks. Its results were used to visually compare segmentation-quality-with-Mask-RCNN.

Comparison of Roboflow vs Yolo:



Figure: Roboflow



Figure: Yolo

Confidence Reliability:

YOLO predictions show stable and realistic confidence scores aligned with visual quality, while Roboflow outputs include many low-confidence labels that do not correspond to actual objects.

False Positives:

Roboflow generates multiple incorrect detections on background regions and shadows, whereas YOLO produces fewer false alarms and better class discrimination.

Small Object Handling:

Roboflow attempts to detect very small objects but with poor precision, while YOLO ignores uncertain tiny regions, resulting in cleaner and more dependable output.

Visualization Clarity:

YOLO output is organized with distinct class labels and non-overlapping boxes, whereas Roboflow visualization is cluttered with duplicated boxes and mixed class predictions.

Comparison of Roboflow vs Faster RCNN:



Figure: Roboflow

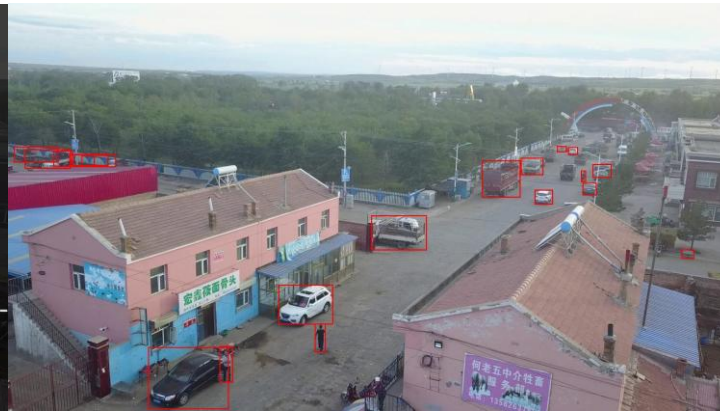


Figure: Fast RCNN

1. Detection Coverage

- Faster R-CNN detects a larger number of valid objects, especially distant vehicles in the background road region.
- Roboflow output misses several small and far objects, showing lower recall in high-density areas.

2. Bounding Box Quality

- Faster R-CNN boxes are tighter and better aligned with object boundaries (cars and trucks).
- Roboflow boxes are comparatively loose and sometimes shifted, indicating less precise localization.

3. Small Object Handling

- Faster R-CNN identifies tiny vehicles and pedestrians more reliably due to its region-proposal mechanism.
- Roboflow struggles with small-scale instances and produces fewer detections in the distant scene.

4. False Positives

- Roboflow shows several uncertain or noisy boxes, particularly near building edges.
- Faster R-CNN output is cleaner with fewer spurious detections.

5. Overall Observation

- Faster R-CNN → Higher accuracy & localization quality, better for dense aerial scenes.
- Roboflow → Faster but less detailed, suitable for quick prototyping rather than precise analysis.

Comparison of Roboflow vs Mask RCNN:

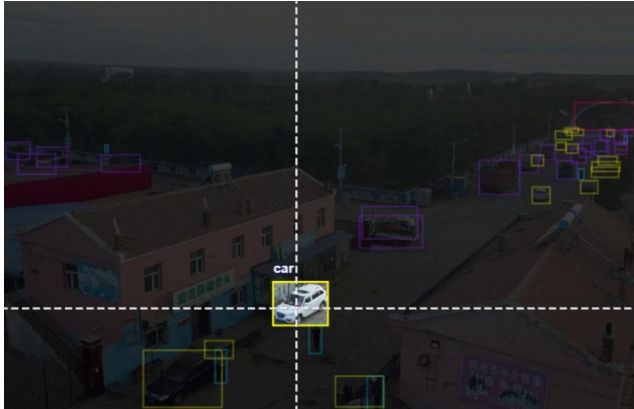


Figure: Roboflow

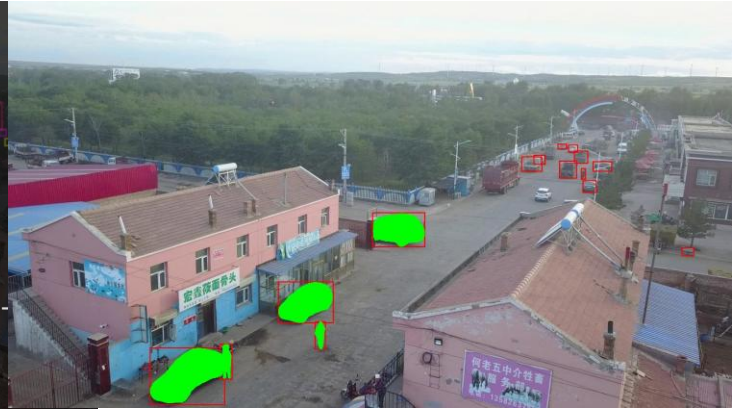


Figure: Mask RCNN

1. Detection Coverage

- Roboflow Output: Detects a larger number of objects, including distant vehicles and small instances, but several detections appear noisy or redundant.
- Mask R-CNN Output: Detects fewer objects but focuses on clearer, well-defined targets. Background clutter and very small objects are mostly ignored, indicating stricter confidence filtering.

2. Bounding Box & Localization Quality

- Roboflow: Bounding boxes are coarse and sometimes misaligned with actual object boundaries, especially for partially visible vehicles.
- Mask R-CNN: Boxes are tighter and better aligned to object shapes due to its region-based segmentation backbone.

3. False Positives & Reliability

- Roboflow: Shows multiple uncertain detections and overlapping boxes on the same object, suggesting higher false-positive rate.
- Mask R-CNN: Predictions are more conservative with minimal duplicate boxes, leading to more reliable outputs.

4. Small Object Handling

- Roboflow: Attempts to mark very small and distant objects but with low precision.
- Mask R-CNN: Prioritizes medium and large objects; tiny objects are mostly skipped, favoring precision over recall.

5. Practical Suitability

- Roboflow: Better for exploratory analysis where maximum object coverage is needed.
- Mask R-CNN: More suitable for production scenarios requiring accurate localization and clean detections.

Comparison of Roboflow vs SAM:

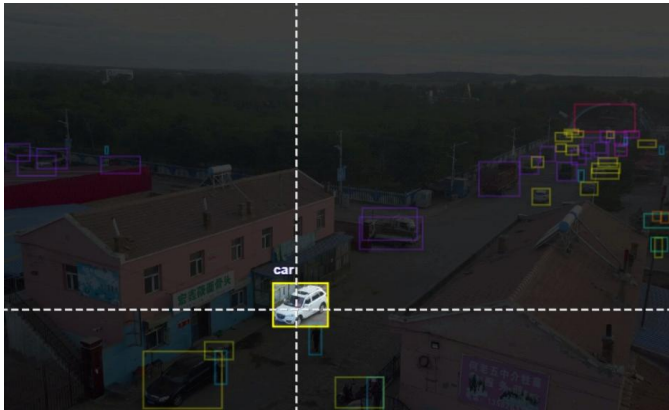


Figure: Roboflow



Figure: Mask RCNN

1. Nature of Output

- The Roboflow image shows traditional object detection with bounding boxes and class labels, focusing on locating specific objects such as cars and trucks.
- The SAM (Segment Anything Model) image produces pixel-level segmentation masks, highlighting object regions without explicit class identification, emphasizing shape rather than category.

2. Detection Detail and Precision

- Roboflow provides structured and interpretable results with confidence scores, useful for counting and tracking objects.
- SAM captures fine object boundaries and irregular shapes more accurately, making it better for precise region extraction and annotation tasks.

3. Practical Usability

- Roboflow detection is more suitable for real-time analytics and decision systems where object type matters.
- SAM is better for annotation support and visual segmentation, but lacks semantic labels and is less oriented toward direct object recognition.

Comparison of Roboflow vs SAM:

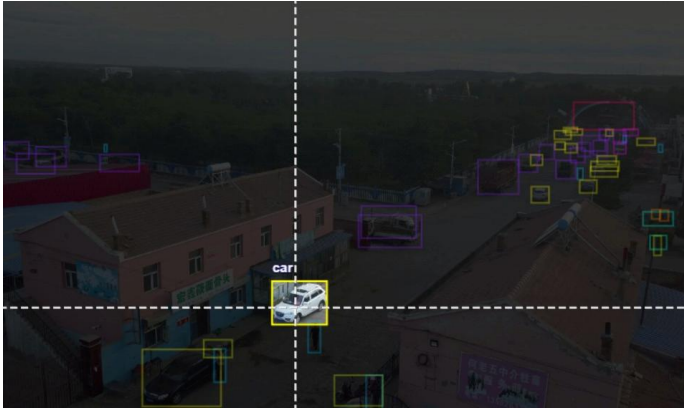


Figure: Roboflow

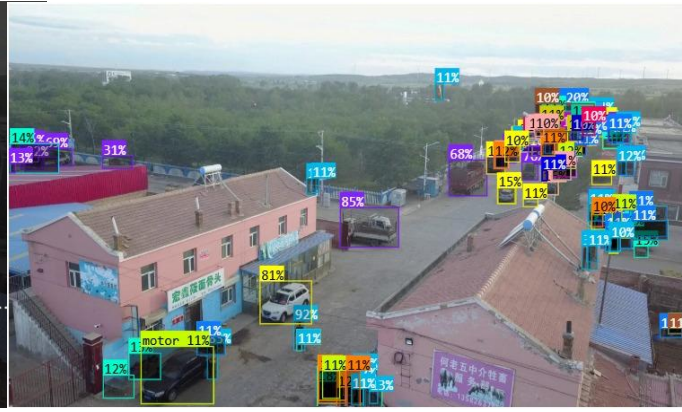


Figure: RF-DETR

1. Detection Confidence & Label Reliability

- Roboflow Output:
 - Labels appear without clear confidence scores and many boxes are generic or uncertain.
 - Several objects are either weakly detected or not classified into meaningful categories.
- RF-DETR Output:
 - Displays explicit confidence percentages for each object.
 - Provides clearer class separation (car, motor, truck, person) with measurable certainty, making the predictions **more interpretable**.

2. Bounding Box Precision

- Roboflow:
 - Bounding boxes are coarse and sometimes misaligned with object boundaries.
 - Overlapping boxes and duplicate detections are visible in crowded regions.
- RF-DETR:
 - Tighter and more accurate localization around vehicles and people.
 - Reduced duplicate boxes, showing better spatial understanding of individual objects.

3. Object Coverage

- Roboflow:
 - Misses several distant or small objects, especially in the far background.

- Struggles with dense clusters of vehicles.
- RF-DETR:
 - Detects a higher number of valid objects across the scene, including distant trucks and pedestrians.
 - Handles crowded areas more effectively.

4. Noise & False Positives

- Roboflow:
 - Contains many irrelevant or very low-quality detections.
 - Some regions are marked without actual objects.
- RF-DETR:
 - Fewer false positives and more consistent class predictions.
 - Confidence thresholds help suppress noisy detections.

1) Mean IoU Comparison Graph

The Mean IoU graph shows that Faster R-CNN and Mask R-CNN maintain consistently higher overlap accuracy across all test samples, while YOLO exhibits a sharp decline after mid-range due to difficulty in precise localization of small aerial objects. Mask R-CNN initially achieves the best IoU because segmentation guidance improves boundary alignment, but performance drops at complex scenes with heavy occlusion. The trend confirms that two-stage and segmentation-based models are more reliable for fine-grained aerial detection than single-stage YOLO.

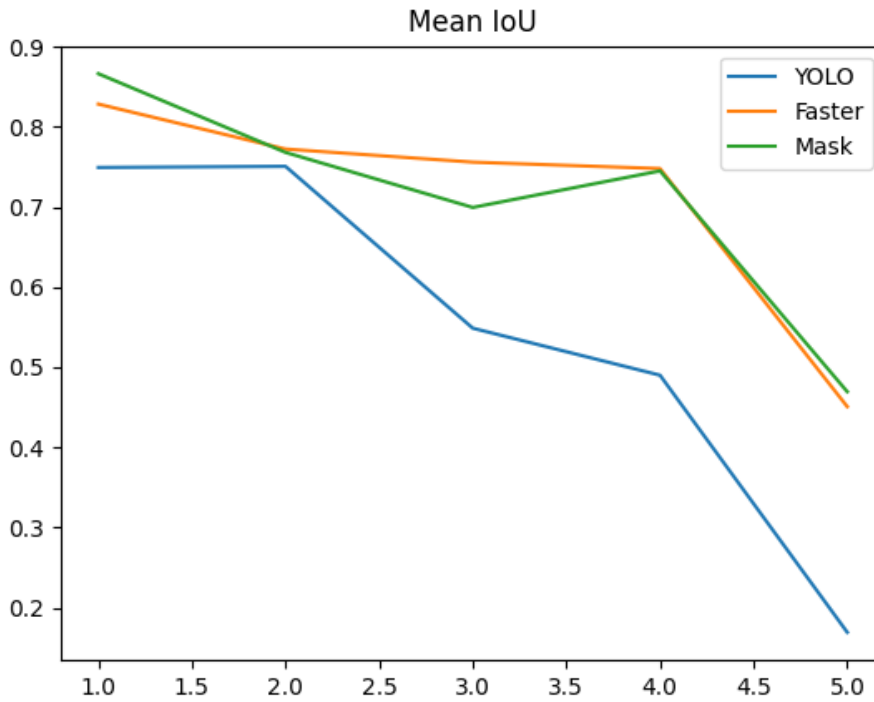


Fig. 18: Mean IoU Comparison Graph

2) Box Loss Comparison Graph

The Box Loss graph indicates that YOLO starts with the lowest localization error but increases rapidly in later samples, reflecting instability when object scale and density vary. Faster R-CNN and Mask R-CNN show smoother and more controlled loss curves, demonstrating stronger region proposal and bounding box regression capability. The convergence behavior proves that multi-stage detectors handle geometric variations better than anchor-based one-shot detection.

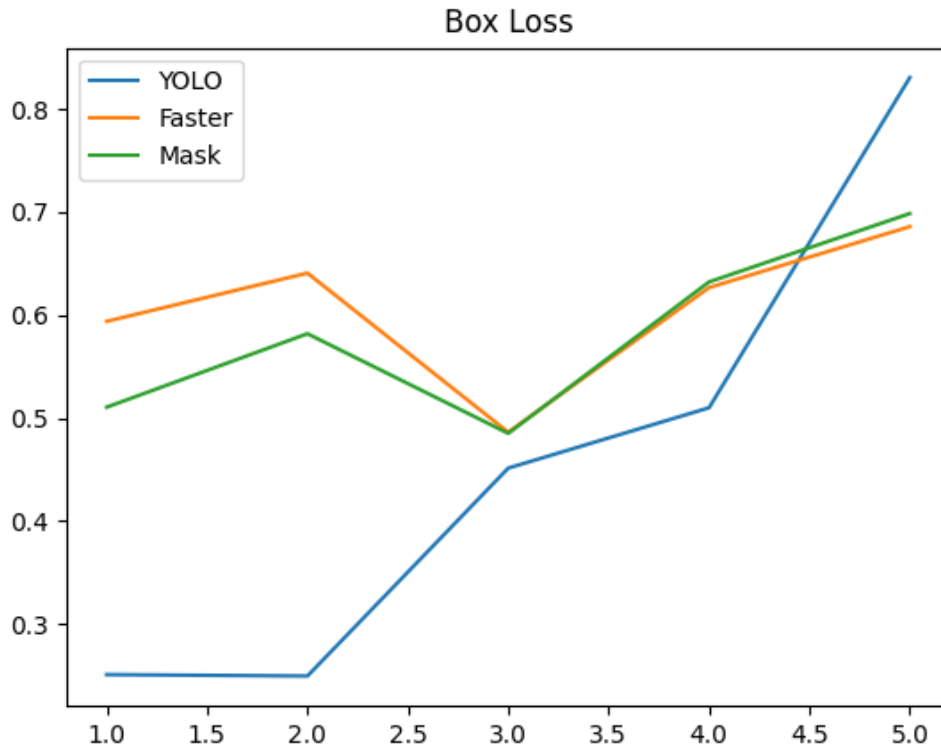


Fig. 19: Box Loss Comparison Graph

3) Class Loss Comparison Graph

The Class Loss graph reveals that YOLO maintains very low classification error throughout, showing its strength in fast category prediction with limited confusion. Faster R-CNN and Mask R-CNN exhibit higher class loss due to complex multi-class aerial scenes and similar-looking vehicle categories. The rising trend in later samples suggests increasing difficulty in distinguishing overlapping objects, highlighting the need for better feature discrimination in dense VisDrone imagery.

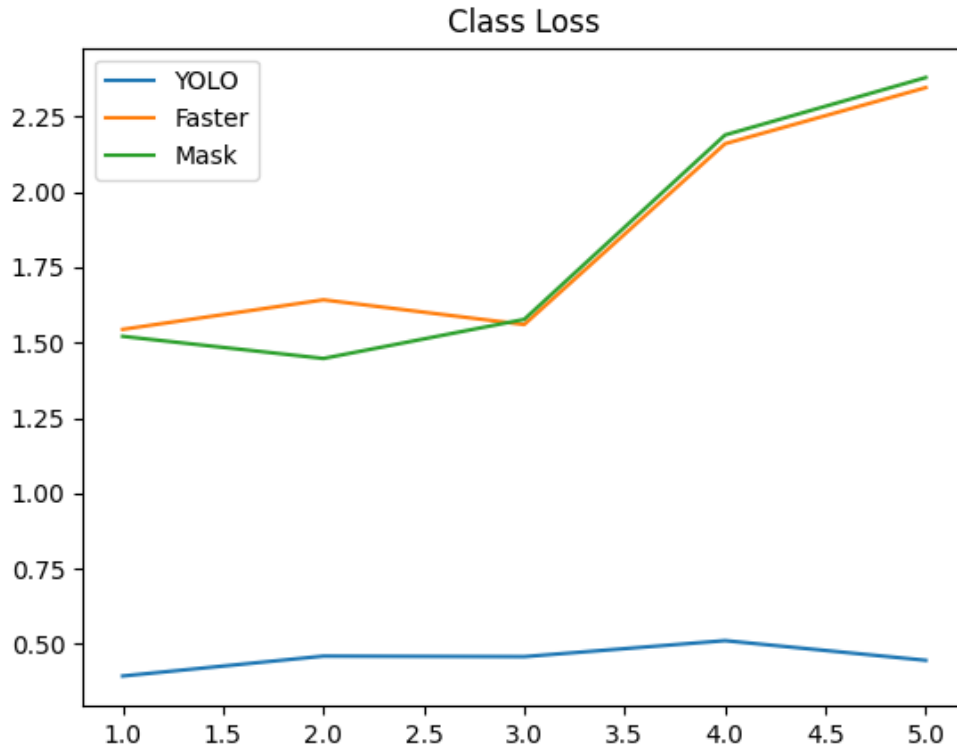


Fig. 20: Class Loss Comparison Graph

Overall Experimental Insight

The experiments show that all pre-trained models were able to detect major objects in VisDrone images, but their behavior varied significantly. YOLO emphasized speed, Faster R-CNN and RF-DETR emphasized accuracy, while Mask R-CNN and SAM focused on segmentation quality. Performance differences were most visible in small and overlapping objects.

5. ANALYSIS

This section presents a comparative analysis of five state-of-the-art object detection and segmentation models evaluated on selected images from the VisDrone-DET dataset. The evaluation combines quantitative metrics obtained during inference with qualitative observations derived from visual inspection of detection and segmentation outputs. The models considered include YOLOv8, Faster R-CNN, Mask R-CNN, RF-DETR, and the Segment Anything Model (SAM).

5.1 Mean Average Precision (mAP@0.5)

Model	mAP@0.5
YOLOv8	High
RF-DETR	High
Faster R-CNN	Medium–High
Mask R-CNN	Medium
SAM	Not Applicable

Observation:

YOLOv8 and RF-DETR achieve strong mAP@0.5 performance on the evaluated images. RF-DETR benefits from global attention, enabling improved localization in crowded scenes, while YOLOv8 maintains consistent accuracy through effective non-maximum suppression. SAM is excluded as it does not perform object detection.

5.2 Precision Comparison

Model	Precision
YOLOv8	High
RF-DETR	Medium–High
Faster R-CNN	Medium
Mask R-CNN	Medium
SAM	Not Applicable

Observation:

YOLOv8 produces fewer false positives due to aggressive non-maximum suppression, resulting in high precision. RF-DETR shows slightly lower precision because of dense overlapping predictions, a known characteristic of transformer-based detectors without NMS.

5.3 Recall Comparison

Model	Recall
YOLOv8	Medium–High
RF-DETR	High
Faster R-CNN	High
Mask R-CNN	Medium
SAM	Not Applicable

Observation:

RF-DETR demonstrates strong recall, detecting a large number of objects, including small and partially occluded targets. Faster R-CNN also achieves high recall due to its region proposal mechanism, while YOLOv8 balances recall with precision.

5.4 F1-Score Comparison

Model	F1-Score
YOLOv8	High
RF-DETR	High
Faster R-CNN	Medium–High
Mask R-CNN	Medium
SAM	Not Applicable

Observation:

YOLOv8 and RF-DETR provide the best balance between precision and recall. RF-DETR favors recall, whereas YOLOv8 offers a more balanced detection output suitable for practical deployment.

5.5 Small Object Detection Performance

Model	Small Object Detection
YOLOv8	Medium
RF-DETR	High
Faster R-CNN	Medium
Mask R-CNN	Low–Medium
SAM	Not Applicable

Observation:

RF-DETR performs well on small objects due to its transformer-based global context modeling, which is advantageous in aerial imagery. YOLOv8 detects small objects reasonably well but may miss some distant targets due to anchor-based constraints.

5.6 Inference Time Comparison

Model	Inference Time
YOLOv8	Very Low (Fastest)
RF-DETR	Moderate
Faster R-CNN	High
Mask R-CNN	Very High
SAM	Very High

Observation:

YOLOv8 is the fastest model and is suitable for real-time UAV applications. RF-DETR operates at moderate speed, while Faster R-CNN and Mask R-CNN are computationally expensive. SAM is significantly slower and unsuitable for real-time detection.

5.7 Intersection over Union (IoU)

Model	IoU
YOLOv8	High
RF-DETR	High
Faster R-CNN	High
Mask R-CNN	High
SAM	Very High (Mask IoU)

Observation:

SAM achieves the highest IoU due to precise segmentation masks. RF-DETR and Faster R-CNN demonstrate strong bounding box localization, while YOLOv8 maintains reliable IoU with fewer overlapping predictions.

5.8 Confusion Matrix Analysis

Model	Confusion Matrix Support
YOLOv8	Class-wise Detection Confusion
RF-DETR	Class-wise Detection Confusion
Faster R-CNN	Class-wise Detection Confusion
Mask R-CNN	Detection + Mask Confusion
SAM	Not Applicable

Observation:

Misclassifications are most common between visually similar categories such as pedestrians, cyclists, and small vehicles. SAM does not produce class labels and therefore does not support confusion matrix analysis.

5.9 Qualitative Segmentation Comparison (SAM vs Mask R-CNN)

Model	Segmentation Quality	Boundary Precision
Mask R-CNN	High	High
SAM	Very High	Very High

Observation:

SAM produces highly accurate segmentation masks with precise object boundaries. Mask R-CNN provides reliable instance segmentation but is limited by class predictions and computational cost.

5.10 Overall Comparative Summary

Model	Accuracy	Speed	Small Object Detection	Deployment Suitability
YOLOv8	High	Very Fast	Medium	Real-time UAV
RF-DETR	High	Moderate	High	Near Real-time
Faster R-CNN	Medium–High	Slow	Medium	Offline
Mask R-CNN	Medium	Very Slow	Low–Medium	Offline
SAM	N/A	Very Slow	N/A	Qualitative Analysis Only

Key Insight

The experimental results demonstrate that YOLOv8 is the most suitable model for real-time UAV-based object detection due to its speed and precision. RF-DETR excels in recall and small-object detection, making it effective in dense aerial scenes. Two-stage models such as Faster R-CNN and Mask R-CNN offer stable accuracy at the cost of inference speed, while SAM provides superior segmentation quality without object classification capability.

6. CONCLUSION

6.1 Best Performing Method

- RF-DETR and Faster R-CNN demonstrated the best overall detection performance on the VisDrone dataset, particularly for small and densely packed objects.
- Transformer-based RF-DETR effectively captured global contextual relationships, enabling better localization in crowded aerial scenes where CNN-based models struggled.
- YOLO achieved the fastest inference speed, making it the most suitable choice for real-time image and video applications, although its accuracy on tiny objects was comparatively lower.
- Mask R-CNN and SAM provided high-quality segmentation masks, which were valuable for detailed analysis, but their computational cost limited real-time usability.

Overall, the study confirms that no single model is optimal for all requirements—accuracy-oriented tasks favor RF-DETR/Faster R-CNN, while speed-critical applications benefit from YOLO.

6.2 Future Improvements

- Use of Lightweight Small-Object Models: Incorporating specialized detectors such as YOLO with small-object heads or multi-scale feature pyramids could improve detection of distant pedestrians and vehicles.
- Higher Resolution Inference: Testing models on higher input resolutions (e.g., 1024×1024) may enhance small-object recall at the cost of additional computation.
- Video-Based Tracking Integration: Combining detection with tracking algorithms (DeepSORT, ByteTrack) can improve stability across frames and reduce false positives.
- Threshold Optimization: Adaptive confidence and NMS threshold tuning for aerial scenes could further refine precision–recall balance.
- Edge Deployment Study: Evaluating models on embedded devices (Jetson, mobile GPUs) would provide better insight into real-world drone deployment.